

Can You Trust What a Model's Activations Tell You?

DataForge 2026 · Pathway Track — required blog post (topic distinct from the main track submission)

Chain-of-thought used to be a convenient accident of how we trained language models: ask for reasoning, get tokens, read the tokens. That convenience is disappearing. Newer architectures reason inside a continuous latent space instead of writing every step down in words, and a handful of production systems now do this by design rather than by omission. That shift raises a question worth taking seriously: if a model never writes its reasoning down, can you still tell what it did — and more importantly, can you tell whether what you're looking at actually caused the answer?

The clearest example of the shift is Coconut (Hao et al., 2024), which trains a model to feed its own last hidden state back in as the next input, instead of decoding a token first. The model keeps reasoning in vector space across several steps before it ever produces language. This isn't a hypothetical concern about future systems — Pathway's BDH-CQ, built on the Dragon Hatchling architecture, is explicitly designed to reason from demonstrations without producing a written chain of thought at all. The tool researchers have leaned on for years — reading the words a model outputs while it thinks — simply doesn't apply to a system like this.

The obvious fallback is to read the internal state directly: train a probe to decode a concept from a layer's activations, and if it decodes well, call that concept "present" or "used." This is where the falsifiable claim in this post lives: probe accuracy is not evidence about whether that information actually caused the model's output. A 2026 paper by Aswal et al., "Observable Patterns Are Not Explanations," tests this directly on two latent reasoning models, Coconut and CODI. Prior work had pointed to observable patterns in their latent states — things like BFS-like search frontiers or decodable arithmetic — as evidence the models were genuinely performing the reasoning mechanism they were designed for. Aswal et al. ran matched control models that lack that proposed mechanism entirely, and found the same observable patterns showed up in the controls too. Using causal interventions, they then showed that a latent "thought"'s actual influence on the model's output is graded rather than binary, and concentrated in specific low-rank directions of the activation space rather than distributed evenly — meaning the parts of the state that look interesting to a probe are not reliably the parts doing the causal work.

The mechanism is worth stating precisely, because it's not mysterious once named: a probe only proves correlation between an activation direction and a labeled concept, and a high-dimensional latent space has room for many such correlated-but-inert directions to exist alongside whatever the model actually computes with. Decodability tells you a signal is

present somewhere in the geometry. It does not tell you the forward pass depended on that signal to produce its answer.

A second 2026 paper, by Baherwani, Goldstein, and Panda, makes the same point from the output side instead of the activation side, with harder numbers attached. Testing 13 frontier language models on synthetic reasoning tasks, they found that semantically meaningless "filler" tokens — text with no relevant content — let several models perform hidden computation that boosted task accuracy by up to 13 percentage points. In one experiment, filler tokens let Claude Opus 4.5 satisfy a hidden modular-arithmetic constraint on the side, without any drop in accuracy on its main task. None of that computation was visible by reading the output tokens. That's the failure case this whole post is built around: the visible trail (tokens, or a probe's best-decoded concept) and the causal trail (what the model actually used) can come apart, in either direction, with no warning sign to tell you they have.

The honest limitation here is that nobody has a cheap, general test for "this activation direction was causally load-bearing" that scales the way probing does. Causal intervention is the right tool in principle — it's what both papers above actually used — but it requires already having a hypothesis about what to intervene on, which is exactly the thing you're usually trying to discover in the first place.

My own view, separated from the evidence above: as more systems adopt latent or token-free reasoning for efficiency reasons — BDH-CQ among them — the interpretability community's reliance on probe accuracy as a proxy for understanding is going to get load-bearing in a way it currently isn't. That's a judgment about where the field's incentives are pointing, not something either paper claims outright. The papers show the gap exists. What we build to close it is still an open problem.